

Optimal Chemo Treatment Durations for Jen

The code for a_2 sums a geometric series of powers of $1/2$ (with negligible tiny corrections), so one finds that for finite n

$$a_2(n) = 2 \sum_{k=3}^{n+2} \frac{1}{2^k} + \frac{1}{2} + (1/4)^n \approx 1 - 2^{-(n+1)} + 2^{-2n},$$

and as $n \rightarrow \infty$ this tends to 1 from below (e.g. $1/2 + 1/4 + 1/8 + \dots = 1$ ¹). In other words, $a_2 < 1$ but extremely close to 1. Thus in practice $a_2 \approx 1$ (computably defined by the code) and in fact $a_2 < a_1 = 1$.

Under the constraint $a_1 x_1 + a_2 x_2 = 1$ (total dose = 1), we have $x_1 = 1 - a_2 x_2$. The total time is

$$T = x_1 + x_2 = 1 + x_2(1 - a_2).$$

Since $1 - a_2 > 0$, T increases with x_2 . Therefore the minimum occurs at $x_2 = 0$ (the “fastest” therapy already achieves the dose). Hence the optimal solution is $x_2 = 0$, $x_1 = 1$. (Equivalently, if a_2 had been larger than 1 the minimum would be at $x_1 = 0$, $x_2 = 1/a_2$.)

In summary, taking a_2 as defined by the code (so $a_2 < 1$), the time-minimising schedule is to use only the first therapy. Numerically, to two-decimal accuracy one can take

$$x_1 \approx 1.00, \quad x_2 \approx 0.00,$$

since $x_1 = 1$, $x_2 = 0$ satisfies $a_1 x_1 + a_2 x_2 = 1$ and any $x_2 > 0$ would increase $x_1 + x_2$. This gives the minimal total time (about 1 unit) to the required precision ¹ ².

Answer: $x_1 \approx 1.00$, $x_2 \approx 0.00$.

¹ ² Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series